



What is Amazon VPC?

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Amazon VPC concepts

Amazon VPC is the networking layer for Amazon EC2.

The following are the key concepts for VPCs:

- Virtual private cloud (VPC) — A virtual network dedicated to your AWS account.
- Subnet — A range of IP addresses in your VPC.
- Route table — A set of rules, called routes, that are used to determine where network traffic is directed.
- Internet gateway — A gateway that you attach to your VPC to enable communication between resources in your VPC and the internet.
- VPC endpoint — Enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. Instances in your VPC do not require public IP addresses to communicate with resources in the service. Traffic between your VPC and the other service does not leave the Amazon network.
- CIDR block — Classless Inter-Domain Routing. An internet protocol address allocation and route aggregation methodology.

Accessing Amazon VPC

You can create, access, and manage your VPCs using any of the following interfaces:

- **AWS Management Console** — Provides a web interface that you can use to access your VPCs.
- **AWS Command Line Interface (AWS CLI)** — Provides commands for a broad set of AWS services, including Amazon VPC, and is supported on Windows, Mac, and Linux.
- **AWS SDKs** — Provides language-specific APIs and takes care of many of the connection details, such as calculating signatures, handling request retries, and error handling.
- **Query API** — Provides low-level API actions that you call using HTTPS requests. Using the Query API is the most direct way to access Amazon VPC, but it requires that your application handle low-level details such as generating the hash to sign the request, and error handling.

Pricing for Amazon VPC

There's no additional charge for using a VPC. There are charges for the following VPC components: Site-to-Site VPN connection, PrivateLink, Traffic Mirroring, and a NAT gateway.

PCI DSS compliance

Amazon VPC supports the processing, storage, and transmission of credit card data by a merchant or service provider, and has been validated as being compliant with Payment Card Industry (PCI) Data Security Standard (DSS).



How Amazon VPC works

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Amazon Virtual Private Cloud (Amazon VPC) enables you to launch AWS resources into a virtual network that you've defined. This virtual network closely resembles a traditional network that you'd operate in your own data center, with the benefits of using the scalable infrastructure of AWS.

Amazon VPC is the networking layer for Amazon EC2.

VPCs and subnets

A virtual private cloud (VPC) is a virtual network dedicated to your AWS account. It is logically isolated from other virtual networks in the AWS Cloud. You can launch your AWS resources, such as Amazon EC2 instances, into your VPC. You can specify an IP address range for the VPC, add subnets, associate security groups, and configure route tables.

A subnet is a range of IP addresses in your VPC. You can launch AWS resources into a specified subnet. Use a public subnet for resources that must be connected to the internet, and a private subnet for resources that won't be connected to the internet .

To protect the AWS resources in each subnet, you can use multiple layers of security, including security groups and network access control lists (ACL).

You can optionally associate an IPv6 CIDR block with your VPC, and assign IPv6 addresses to the instances in your VPC.

Default and nondefault VPCs

If your account was created after 2013-12-04, it comes with a default VPC that has a default subnet in each Availability Zone. A default VPC has the benefits of the advanced features provided by EC2-VPC, and is ready for you to use. If you have a default VPC and don't specify a subnet when you launch an instance, the instance is launched into your default VPC. You can launch instances into your default VPC without needing to know anything about Amazon VPC.

You can also create your own VPC, and configure it as you need. This is known as a nondefault VPC. Subnets that you create in your nondefault VPC and additional subnets that you create in your default VPC are called nondefault subnets.

Route tables

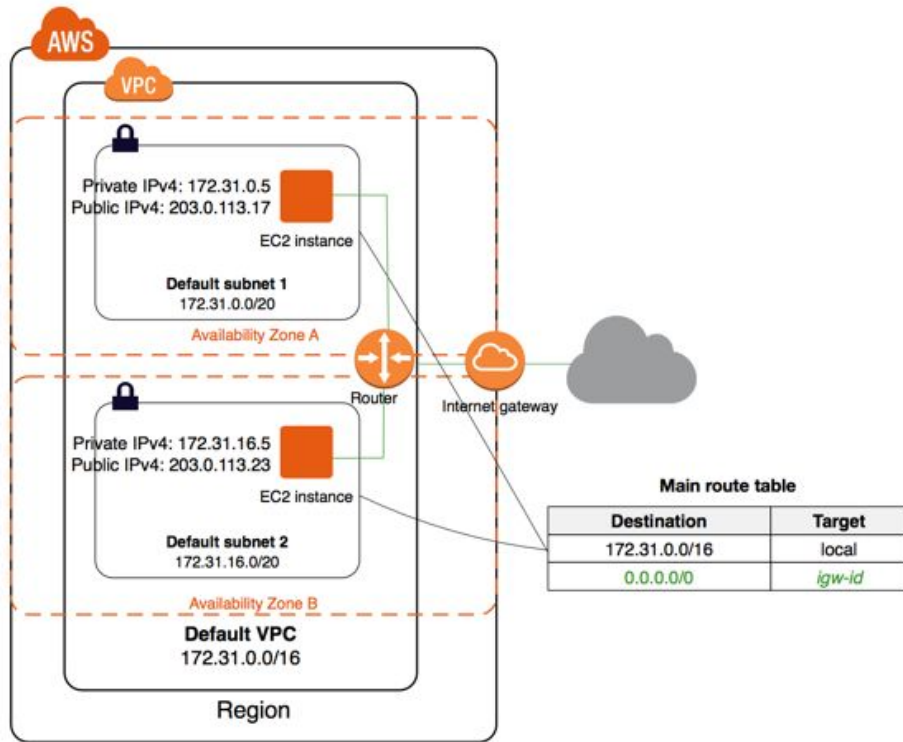
A route table contains a set of rules, called routes , that are used to determine where network traffic from your VPC is directed. You can explicitly associate a subnet with a particular route table. Otherwise, the subnet is implicitly associated with the main route table.

Each route in a route table specifies the range of IP addresses where you want the traffic to go (the destination) and the gateway, network interface, or connection through which to send the traffic (the target).

Accessing the internet

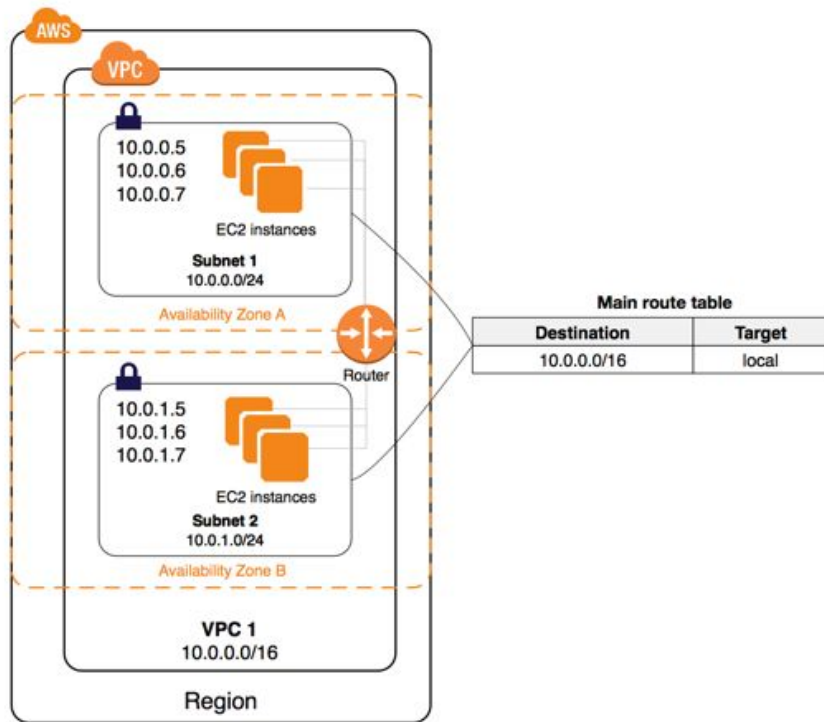
You control how the instances that you launch into a VPC access resources outside the VPC.

Your default VPC includes an internet gateway, and each default subnet is a public subnet. Each instance that you launch into a default subnet has a private IPv4 address and a public IPv4 address. These instances can communicate with the internet through the internet gateway. An internet gateway enables your instances to connect to the internet through the Amazon EC2 network edge.



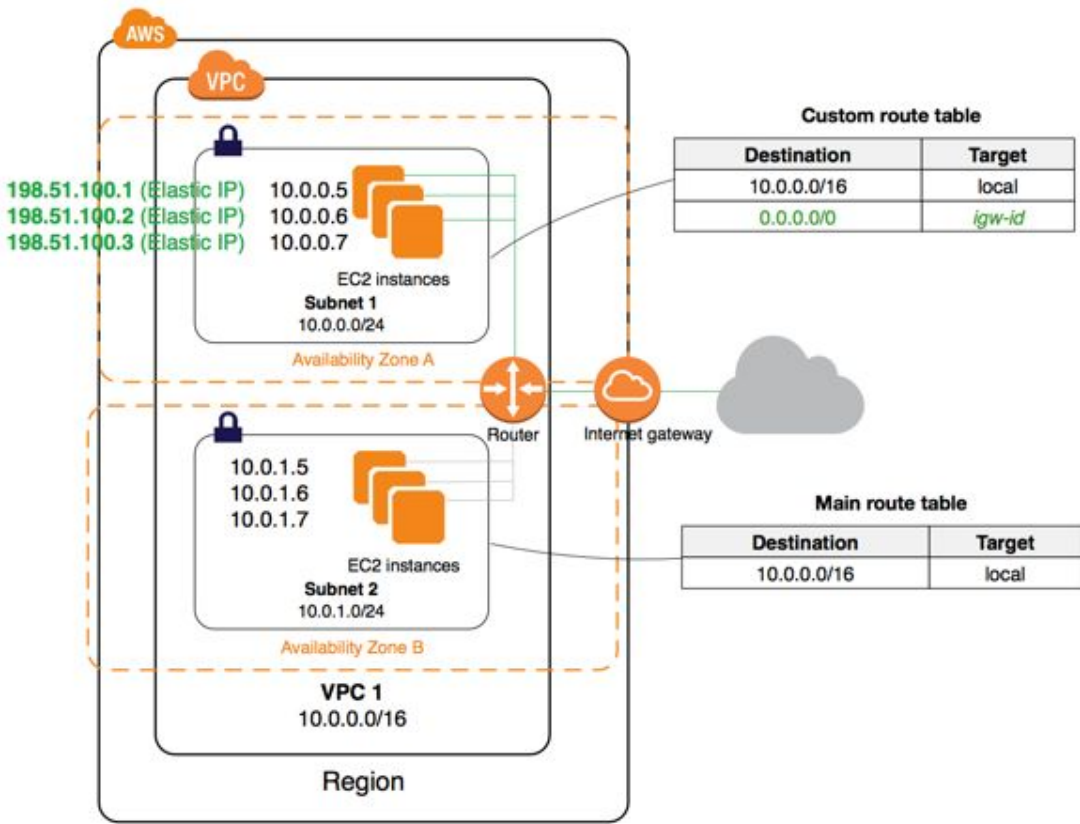
Accessing the internet

By default, each instance that you launch into a nondefault subnet has a private IPv4 address, but no public IPv4 address, unless you specifically assign one at launch, or you modify the subnet's public IP address attribute. These instances can communicate with each other, but can't access the internet.



Accessing the internet

You can enable internet access for an instance launched into a nondefault subnet by attaching an internet gateway to its VPC (if its VPC is not a default VPC) and associating an Elastic IP address with the instance.



Accessing the internet

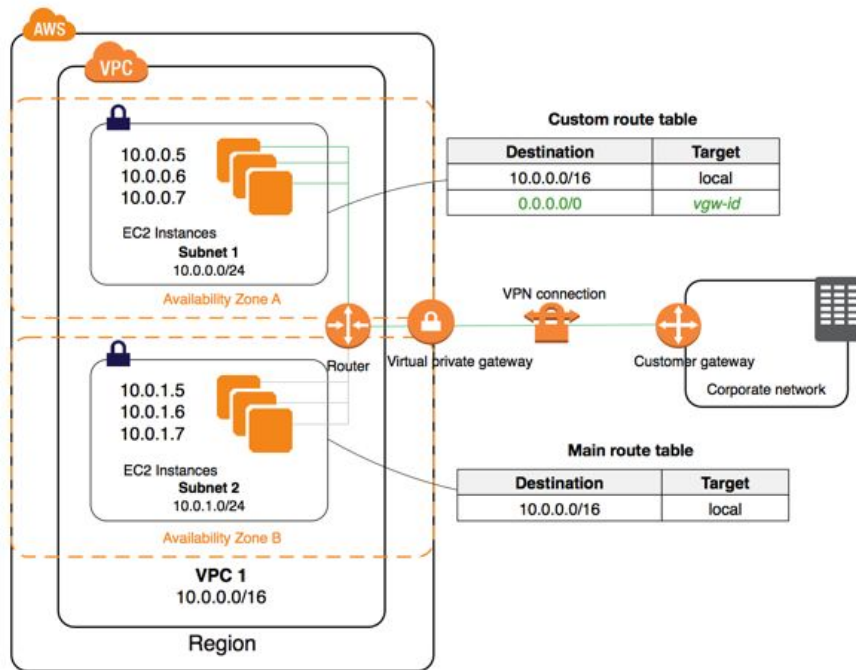
Alternatively, to allow an instance in your VPC to initiate outbound connections to the internet but prevent unsolicited inbound connections from the internet, you can use a network address translation (NAT) device for IPv4 traffic. NAT maps multiple private IPv4 addresses to a single public IPv4 address. A NAT device has an Elastic IP address and is connected to the internet through an internet gateway. You can connect an instance in a private subnet to the internet through the NAT device, which routes traffic from the instance to the internet gateway, and routes any responses to the instance.

If you associate an IPv6 CIDR block with your VPC and assign IPv6 addresses to your instances, instances can connect to the internet over IPv6 through an internet gateway. Alternatively, instances can initiate outbound connections to the internet over IPv6 using an egress-only internet gateway. IPv6 traffic is separate from IPv4 traffic; your route tables must include separate routes for IPv6 traffic.

Accessing a corporate or home network

You can optionally connect your VPC to your own corporate data center using an IPsec AWS Site-to-Site VPN connection, making the AWS Cloud an extension of your data center.

A Site-to-Site VPN connection consists of two VPN tunnels between a virtual private gateway or transit gateway on the AWS side, and a customer gateway located in your data center. A customer gateway device is a physical device or software appliance that you configure on your side of the Site-to-Site VPN connection.

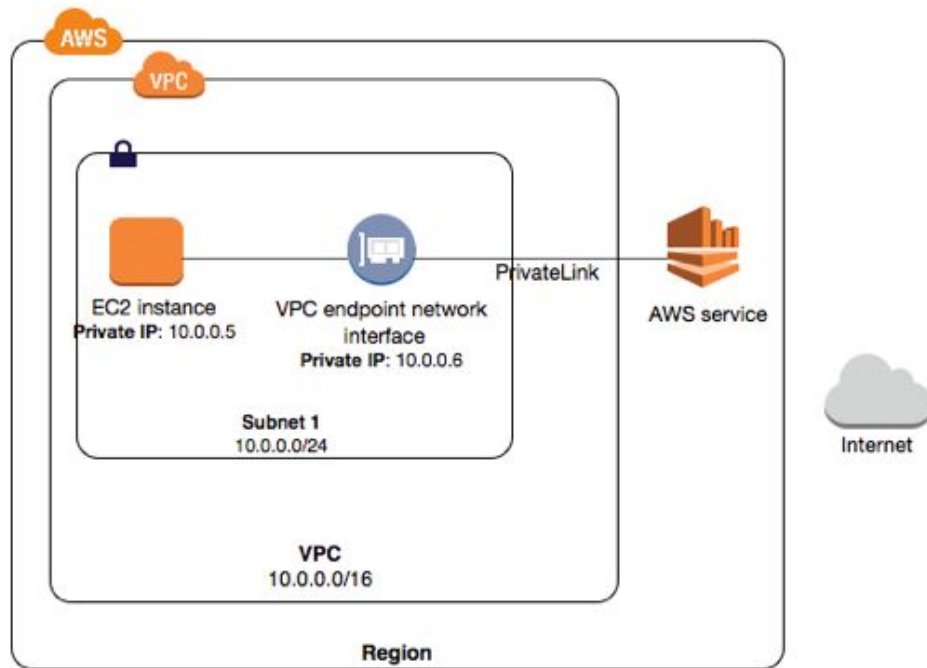


Accessing services through AWS PrivateLink

AWS PrivateLink is a highly available, scalable technology that enables you to privately connect your VPC to supported AWS services, services hosted by other AWS accounts (VPC endpoint services), and supported AWS Marketplace partner services. You do not require an internet gateway, NAT device, public IP address, AWS Direct Connect connection, or AWS Site-to-Site VPN connection to communicate with the service. Traffic between your VPC and the service does not leave the Amazon network.

Accessing services through AWS PrivateLink

To use AWS PrivateLink, create a VPC endpoint for a service in your VPC. You create the type of VPC endpoint required by the supported service. This creates an elastic network interface in your subnet with a private IP address that serves as an entry point for traffic destined to the service. You can create your own AWS PrivateLink-powered service (endpoint service) and enable other AWS customers to access your service.



Connecting VPCs and networks

You can create a VPC peering connection between two VPCs that enables you to route traffic between them privately. Instances in either VPC can communicate with each other as if they are within the same network.

You can also create a transit gateway and use it to interconnect your VPCs and on-premises networks. The transit gateway acts as a Regional virtual router for traffic flowing between its attachments, which can include VPCs, VPN connections, AWS Direct Connect gateways, and transit gateway peering connections.

AWS private global network considerations

AWS provides a high-performance, and low-latency private global network that delivers a secure cloud computing environment to support your networking needs. AWS Regions are connected to multiple Internet Service Providers (ISPs) as well as to a private global network backbone, which provides improved network performance for cross-Region traffic sent by customers.

The following considerations apply:

- Traffic that is in an Availability Zone, or between Availability Zones in all Regions, routes over the AWS private global network.
- Traffic that is between Regions always routes over the AWS private global network, except for China Regions.

Network packet loss can be caused by a number of factors, including network flow collisions, lower level (Layer 2) errors, and other network failures. We engineer and operate our networks to minimize packet loss. We measure packet-loss rate (PLR) across the global backbone that connects the AWS Regions. We operate our backbone network to target a p99 of the hourly PLR of less than 0.0001%.

Supported platforms

The original release of Amazon EC2 supported a single, flat network that's shared with other customers called the EC2-Classic platform. Earlier AWS accounts still support this platform, and can launch instances into either EC2-Classic or a VPC. Accounts created after 2013-12-04 support EC2-VPC only.



Getting started with Amazon VPC

Getting started with Amazon VPC

To get started using Amazon VPC, you can create a nondefault VPC. The following steps describe how to use the Amazon VPC wizard to create a nondefault VPC with a public subnet, which is a subnet that has access to the internet through an internet gateway. You can then launch an instance into the subnet and connect to it.

Step 1: Create the VPC

In this step, you'll use the Amazon VPC wizard in the Amazon VPC console to create a VPC. The wizard performs the following steps for you:

- Creates a VPC with a /16 IPv4 CIDR block (a network with 65,536 private IP addresses).
- Attaches an internet gateway to the VPC.
- Creates a size /24 IPv4 subnet (a range of 256 private IP addresses) in the VPC.
- Creates a custom route table, and associates it with your subnet, so that traffic can flow between the subnet and the internet gateway.

To create a VPC using the Amazon VPC Wizard

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the navigation bar, on the top-right, take note of the AWS Region in which you'll be creating the VPC. Ensure that you continue working in the same Region for the rest of this exercise, as you cannot launch an instance into your VPC from a different Region.
3. In the navigation pane, choose **VPC dashboard**. From the dashboard, choose **Launch VPC Wizard**.
4. Choose **VPC with a Single Public Subnet**, and then choose **Select**.
5. On the configuration page, enter a name for your VPC in the **VPC name** field; for example, my-vpc, and enter a name for your subnet in the **Subnet name** field. This helps you to identify the VPC and subnet in the Amazon VPC console after you've created them. For this exercise, leave the rest of the configuration settings on the page, and choose **Create VPC**.
6. A status window shows the work in progress. When the work completes, choose **OK** to close the status window.
7. The Your VPCs page displays your default VPC and the VPC that you just created. The VPC that you created is a nondefault VPC, therefore the **Default VPC** column displays **No**.

View information about your VPC

After you've created the VPC, you can view information about the subnet, the internet gateway, and the route tables. The VPC that you created has two route tables — a main route table that all VPCs have by default, and a custom route table that was created by the wizard. The custom route table is associated with your subnet, which means that the routes in that table determine how the traffic for the subnet flows. If you add a new subnet to your VPC, it uses the main route table by default.

To view information about your VPC

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the navigation pane, choose **Your VPCs**. Take note of the name and the ID of the VPC that you created (look in the **Name** and **VPC ID** columns). You will use this information to identify the components that are associated with your VPC.
3. In the navigation pane, choose **Subnets**. The console displays the subnet that was created when you created your VPC. You can identify the subnet by its name in **Name** column, or you can use the VPC information that you obtained in the previous step and look in the **VPC** column.

To view information about your VPC

4. In the navigation pane, choose **Internet Gateways**. You can find the internet gateway that's attached to your VPC by looking at the **VPC** column, which displays the ID and the name (if applicable) of the VPC.
5. In the navigation pane, choose **Route Tables**. There are two route tables associated with the VPC. Select the custom route table (the **Main** column displays **No**), and then choose the **Routes** tab to display the route information in the details pane:
 - The first row in the table is the local route, which enables instances within the VPC to communicate. This route is present in every route table by default, and you can't remove it.
 - The second row shows the route that the Amazon VPC wizard added to enable traffic destined for the internet (0.0.0.0/0) to flow from the subnet to the internet gateway.
6. Select the main route table. The main route table has a local route, but no other routes.

Step 2: Launch an instance into your VPC

When you launch an EC2 instance into a VPC, you must specify the subnet in which to launch the instance. In this case, you'll launch an instance into the public subnet of the VPC you created. You'll use the Amazon EC2 launch wizard in the Amazon EC2 console to launch your instance.

To launch an EC2 instance into a VPC

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.
2. In the navigation bar, on the top-right, ensure that you select the same Region in which you created your VPC.
3. From the dashboard, choose Launch Instance.
4. On the first page of the wizard, choose the AMI that you want to use. For this exercise, choose an Amazon Linux AMI or a Windows AMI.

To launch an EC2 instance into a VPC

5. On the **Choose an Instance Type** page, you can select the hardware configuration and size of the instance to launch. By default, the wizard selects the first available instance type based on the AMI you selected. You can leave the default selection, and then choose **Next: Configure Instance Details**.
6. On the **Configure Instance Details** page, select the VPC that you created from the **Network** list, and the subnet from the **Subnet** list. Leave the rest of the default settings, and go through the next pages of the wizard until you get to the **Add Tags** page.
7. On the **Add Tags** page, you can tag your instance with a Name tag; for example Name=MyWebServer. This helps you to identify your instance in the Amazon EC2 console after you've launched it. Choose **Next: Configure Security Group** when you are done.
8. On the **Configure Security Group** page, the wizard automatically defines the launch-wizard-x security group to allow you to connect to your instance. Choose **Review and Launch**.

To launch an EC2 instance into a VPC

1. On the **Review Instance Launch** page, choose **Launch**.
2. In the **Select an existing key pair or create a new key pair** dialog box, you can choose an existing key pair, or create a new one. If you create a new key pair, ensure that you download the file and store it in a secure location. You'll need the contents of the private key to connect to your instance after it's launched.

To launch your instance, select the acknowledgment check box, and then choose **Launch Instances**.

3. On the confirmation page, choose **View Instances** to view your instance on the **Instances** page. Select your instance, and view its details in the **Description** tab. The **Private IPs** field displays the private IP address that's assigned to your instance from the range of IP addresses in your subnet.

Step 3: Assign an Elastic IP address to your instance

In the previous step, you launched your instance into a public subnet — a subnet that has a route to an internet gateway. However, the instance in your subnet also needs a public IPv4 address to be able to communicate with the internet. By default, an instance in a nondefault VPC is not assigned a public IPv4 address. In this step, you'll allocate an Elastic IP address to your account, and then associate it with your instance.

To allocate and assign an Elastic IP address

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the navigation pane, choose **Elastic IPs**.
3. Choose **Allocate new address**, and then **Allocate**.
4. Select the Elastic IP address from the list, choose **Actions**, and then choose **Associate Address**.
5. For **Resource type**, ensure that **Instance** is selected. Choose your instance from the **Instance** list. Choose **Associate** when you're done.

Your instance is now accessible from the internet. You can connect to your instance through its Elastic IP address using SSH or Remote Desktop from your home network.

Step 4: Clean up

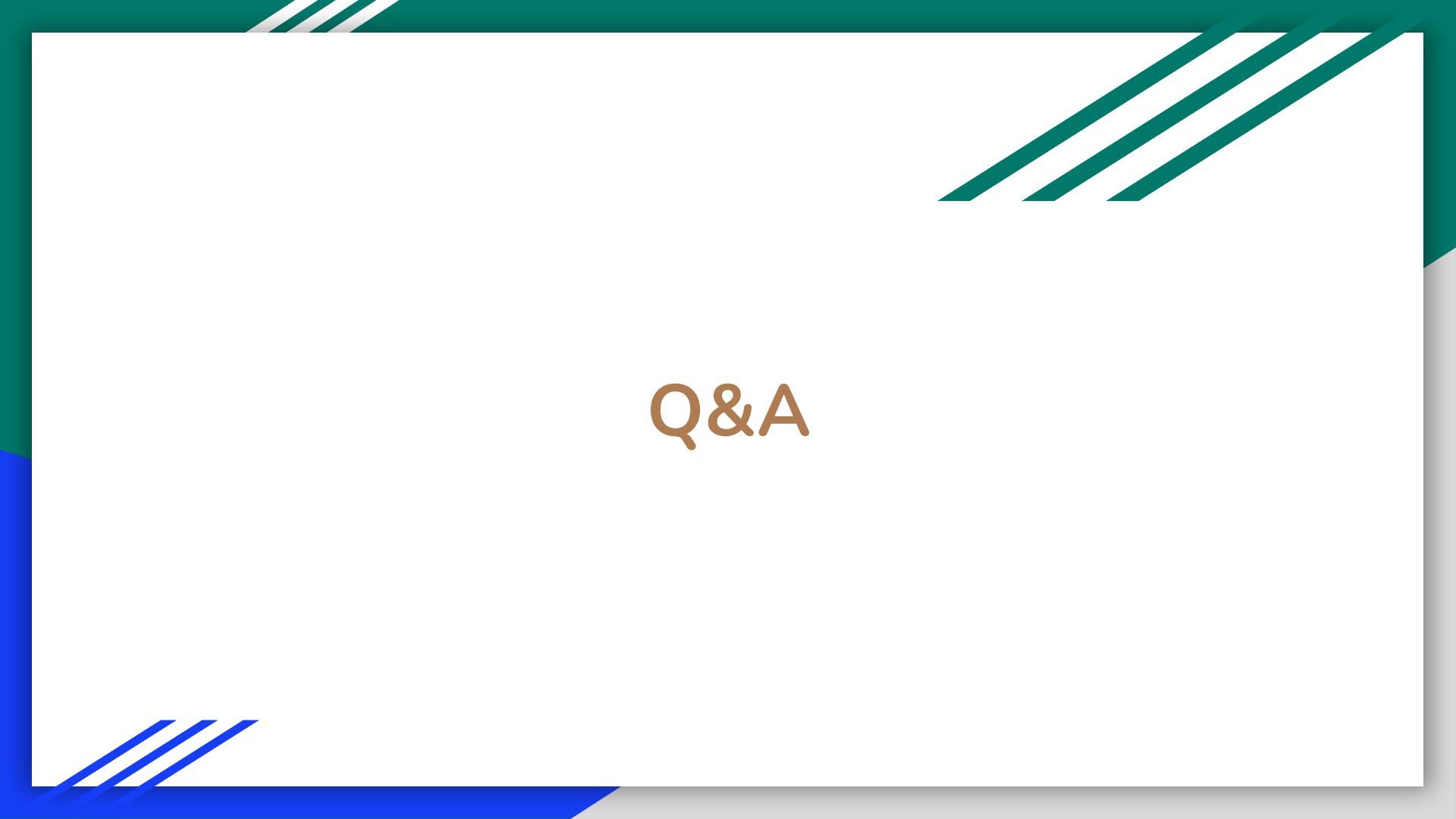
You can choose to continue using your instance in your VPC, or if you do not need the instance, you can terminate it and release its Elastic IP address to avoid incurring charges for them. You can also delete your VPC — note that you are not charged for the VPC and VPC components created in this exercise (such as the subnets and route tables).

Before you can delete a VPC, you must terminate any instances that are running in the VPC. You can then delete the VPC and its components using the VPC console.

Step 4: Clean up

To terminate your instance, release your Elastic IP address, and delete your VPC

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.
2. In the navigation pane, choose **Instances**.
3. Select your instance, choose **Actions**, then **Instance State**, and then select **Terminate**.
4. In the dialog box, expand the **Release attached Elastic IPs** section, and select the check box next to the Elastic IP address. Choose **Yes, Terminate**.
5. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
6. In the navigation pane, choose **Your VPCs**.
7. Select the VPC, choose **Actions**, and then choose **Delete VPC**.
8. When prompted for confirmation, choose **Delete VPC**.



Q&A